

SOFTWARE That Can FLY YOUR UAVs And DRONES

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BRL CAD

BRL-CAD is an open source combinatorial constructive solid geometry (CSG) modelling system. According to its legacy, the US military also used this software to simulate weapon systems for vulnerability and lethality analyses. Interestingly, solid modelling is frequently used in academia and industries apart from defence simulations. This software packs itself with interactive 3D solid geometry editing, high-performance ray-tracing support for rendering and geometric analysis. With network-distributed frame buffer support, image- and signal-processing tools, path-tracing and photon mapping support, the software helps simulate realistic images of events.

As an add-on, there is an embedded scripting interface and libraries for robust high-performance geometric representation and analysis in this tool. Some interesting simulations with this software include radiation dose planning, medical visualisation, computer graphics education, CSG concepts and modelling education, and system performance benchmark testing.

Magic VLSI

Magic is an open source EDA tool that directly competes with other commercial tools in providing features for streamlining product design flow. The simplified design rules and circuit structures take manufacturing and designing process a step ahead in time, while the interactive editor and built-in layout rules, circuits, transistors and efficient routing support intuitively assist in last minute modifications and

Some popular resources

OpenDroneMap. Do you love ariel drone imagery or, to simply put across, looking at the visual perspective provided by drones or unmanned aerial vehicles (UAVs)? OpenDroneMap is that open source toolkit for processing aerial drone imagery that gives you better imagery than typical drone point-and-shoot cameras. It turns simple images into 3D geographic data that can be used in combination with other geographic datasets.

Paparazzi UAV. Paparazzi is a popular open source drone hardware and software project that focuses itself on autopilot systems and ground station software for multicopters/multirotors, fixed-wing, helicopters and hybrid aircraft. This utility was designed with autonomous flight as the primary focus and manual flying as the secondary, and to be made compatible with the ability to control multiple aircraft within the same system. Featuring a dynamic flight plan system defined by mission states, Paparazzi makes it easy to create complex automated missions without the operator or pilot's intervention.

dissolve the complexity involved in the design process. Magic was originally coded to focus on tool command language but, over time, its engineers optimised it to process designs and fabrications.

myNetPCB

If you ever wanted a PCB application that has all the convenient features to support your design, the search would end with myNetPCB. This Java based Windows application helps you create schematic circuits using custom symbols, inserting wires, drawing buses, junctions and connections. The software could be stored on any USB flash drive and launched from the same location, unlike other software that are not portable in a USB. Being a Java application, it enables you to carry your designs anywhere, and plug and play from what has been saved to the utility. The software gives the freedom to import data from XML file format and export the gener-

ated design to extensible markup language (XML), portable network graphics (PNG) or joint photographic experts group (JPG) file format. It contains editing options to load symbols of inductor, LCD, diode, ground, resistor, resonator, transistor or jumper, along with the possibility to create user-defined symbols using various drawing tools, such as rectangles, ellipses, arcs, lines, triangles, pins and labels.

IndigoSCADA

Supervisory Control And Data Acquisition (SCADA) are dedicated software application programs for process control and real-time data gathering from remote locations to control equipment and conditions. With IndigoSCADA you can capture real-time and historical graphical representation of data in your manufacturing unit.

PX4

This open source package of software introduces you to setting up and flying PX4 drones while learning all that there is about flight control and configurations. With this software, learn how to tune a multicopter, sync flight controller with autopilot software and much more. PX4 ground control station is called QGroundControl, bundled into the package, which is an integral part of the autopilot system. QGroundControl runs on Widows, Linux and other platforms, and gets you started with real-time flight information, fully autonomous missions and drone racing. **EFY**

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